Project Rubric (100 points) — Fillable

Course/Program: ____________________ **Project:** ____________________

Instructor: ____________________ **Team Name:** ____________________

Date: ____________________

Category	**Max Points**	**Score (enter)**	**Comments**
Problem Framing & Requirements	10		
Technical Modeling & Correctness	25		
Analysis, Validation & Sanity Checks	20		
Code Quality, Reproducibility & Documentation	15		
Communicating Project Findings	10		
Results, Interpretation & Recommendations	10		
Teamwork & Collaboration	10		
TOTAL	**100**		

Scoring Details

1) Problem Framing & Requirements (10 pts)

- **Excellent (9–10):** Clearly states problem, objectives, constraints, and success criteria; explicitly maps tasks to deliverables
- **Proficient (7–8):** Objectives and constraints are identified; minor gaps in success criteria or deliverable mapping.
- **Developing (4–6):** Problem statement is partial; constraints/criteria not well defined; some deliverables unclear.
- **Insufficient (0–3):** Unclear objective; requirements not addressed.

2) Technical Modeling & Correctness (25 pts)

Assess mathematical formulation, implementation, and correctness relative to each project's core methods.

- **Excellent (22–25):** ____ (instructor defines according to project specifics)
- **Proficient (18–21):** Minor formulation or implementation errors that don't change conclusions; sound parameter choices; units mostly consistent.
- **Developing (12–17):** Noticeable errors (e.g., sign mistakes, unit inconsistencies, missing constraints) that weaken conclusions.
- **Insufficient (0–11):** Fundamental modeling errors; model not executable or not aligned with physical/optimization principles.

3) Analysis, Validation & Sanity Checks (20 pts)

Look for triangulation: hand calculations vs. simulation; sensitivity; bounds/edge cases.

- **Excellent (18–20):**
  - Compares hand calculations with MATLAB results; explores sensitivity; checks bounds/edge cases
  - Quantifies uncertainty or error and explains discrepancies.
  - ____ (instructor defines according to project specifics)
- **Proficient (14–17):** Includes basic cross-checks; touches on sensitivity or edge cases; provides plausible error discussion.
- **Developing (9–13):** Minimal cross-checks; limited sensitivity analysis; inconsistencies not explained.
- **Insufficient (0–8):** No validation; results taken at face value.

4) Code Quality, Reproducibility, & Documentation (15 pts)

Evaluate repository organization, documentation, and ability to re-run.

- **Excellent (14–15):** Clear file organization; README with setup/run steps (if using); code comments; parameters/data documented; figures regenerate from scripts.

- **Proficient (11–13):** Mostly organized; README present (if using); minor gaps (e.g., missing figure regeneration step).
- **Developing (7–10):** Disorganized; sparse documentation; partial reproducibility (manual steps required).
- **Insufficient (0–6):** Hard to run; missing essential files; no documentation.

5) Communicating Project Findings (10 pts)

Consider clarity, correctness, and narrative in report/presentation.

- **Excellent (9–10):** High quality visuals; labeled axes/units; captions link visuals to conclusions; clear, concise writing; logical slide deck.
    - ______________________________________ (instructor defines according to project specifics)
- **Proficient (7–8):** Good visuals; minor labeling or narrative gaps.
- **Developing (4–6):** Basic visuals; cluttered or under labeled; narrative hard to follow.
- **Insufficient (0–3):** Visuals missing or misleading; unclear writing/presentation.

6) Results, Interpretation & Recommendations (10 pts)

Judge whether conclusions are supported and actionable.

- **Excellent (9–10):** Results summarized with evidence; interprets physical/engineering meaning; offers practical next steps/limitations.
    - ______________________________________ (instructor defines according to project specifics)
- **Proficient (7–8):** Conclusions mostly supported; some limitations acknowledged; actionable suggestions.
- **Developing (4–6):** Over-generalized claims; limited linkage to evidence; few or no recommendations.
- **Insufficient (0–3):** Unsupported conclusions; misinterpretation of results.

7) Teamwork & Collaboration (10 pts)

Observe process, equity, and professionalism throughout the project.

- **Excellent (9–10):** Clear role assignment; consistent meeting cadence; responsive communication; equitable workload; constructive peer review; resolves conflicts professionally; timely delivery.
- **Proficient (7–8):** Roles mostly clear; regular updates; minor unevenness in workload or responsiveness.
- **Developing (4–6):** Irregular communication; uneven contribution; limited peer review.
- **Insufficient (0–3):** Poor collaboration; chronic missed deadlines; unresolved conflict; one-person project.